

What Explains the Low Survival Rate of Developing Country Export Flows?

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Successful export growth and diversification require not only entry into new export products and markets but also the survival and growth of export flows. For a cross-country dataset of product-level bilateral export flows, exporting is found to be a perilous activity, especially in low-income countries. Unobserved individual heterogeneity in product-level export flow data prevails even when a wide range of observed country and product characteristics are controlled for. This questions previous studies that used the Cox proportional hazards model to analyze export survival. Following Meyer (1990), a Prentice-Gloeckler (1978) model is estimated, amended with a gamma mixture distribution summarizing unobserved individual heterogeneity. The empirical results confirm the significance of a range of product- as well as country-specific factors in determining the survival of new export flows. Important for policy-making is the finding of the value of learning-by-doing for export survival: experience with exporting the same product to other markets or different products to the same market is found to strongly increase the chance of export survival. A better understanding of such learning effects could substantially improve the effectiveness of export promotion strategies. Export survival, Cox proportional hazard, low-income countries. JEL codes: F10, F14, C41

In a seminal study, Besedes and Prusa (2006b) conclude that for developing countries the key element in achieving faster aggregate export growth is higher survival rates for existing trade flows. Brenton, Pierola, and von Uexkull (2009) show that poorly performing developing countries are not inferior to stronger performing countries in introducing new trade flows but that these countries

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THE WORLD BANK ECONOMIC REVIEW, VOL. 24, NO. 3, pp. 474–499
Advance Access Publication December 14, 2010

doi:10.1093/wber/lhq015

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experience much lower rates of survival. It is these low survival rates that undermine the expansion of export flows.

From a practical policy perspective, it is important to understand not only the factors driving entry into exporting but also the processes that sustain new exports and expand export volume. Are there factors that are amenable to public policy interventions that affect the survival of a particular exporter or export flow? Do these factors impinge particularly on firms in low-income countries?

This article extends the empirical analysis of export survival to provide a basis for answering these questions, building on Besedes and Prusa's (2006b) initial study of the determinants of export flow duration.¹ First, it investigates a novel, more detailed dataset of product-level trade flows comprising a large number of both developing and developed country imports and exports that includes low-income as well as middle- and high-income exporters. Second, it pursues a more rigorous empirical approach that examines the reliability of the continuous time proportional hazards model that has been used unquestioningly to assess the determinants of the survival rates of new export flows (Besedes and Prusa 2006b; Blyde 2008). This model is misspecified in the presence of unobserved heterogeneity, a strong possibility with export data—for example, from the unobserved quality of the management of the underlying exporting firms.

The findings of the empirical analysis provide strong evidence against the validity of the proportional hazards assumption in the Cox model. A possible explanation is the presence of unobserved individual heterogeneity that is not accounted for in the estimation. Given the nature of the trade flow data, which group survival rates into annual observations and have many tied failure rates, a Prentice-Gloeckler (1978) model is used to provide a discrete time equivalent of the Cox proportional hazards model. The model is augmented with a gamma mixture distribution to address unobserved individual heterogeneity (Meyer 1990). The analysis reveals that these methodological improvements are indeed important. The coefficients change strongly in switching from the Cox to the Prentice-Gloeckner model, and the estimated impact on the hazard rate of most variables rises. Augmenting the Prentice-Gloeckner model with a gamma mixture distribution further increases the impact of all variables. For most of the variables of interest, this change is economically significant.

In general, failure to account for unobserved heterogeneity, if it is indeed important, will overestimate the degree of negative duration dependence in the (true) baseline hazard and underestimate the degree of positive duration dependence. Moreover, the proportionate effect of a given regressor on the hazard rate is no longer constant and independent of survival time, and the proportionate response of the hazard to variation in each regressor at any survival time is attenuated. All of this is precisely what is observed in the data.

1. Blyde (2008) provides another application of the basic approach used by Besedes and Prusa (2006b).

The empirical results confirm that cultural and geographic ties between trading partners as well as market size and exporting experience are important to the survival of new export flows. Moreover, the results suggest that exporting experience is product and region specific, particularly for developing country exporters. In line with standard findings in the literature, variables designed to capture search costs as well as cultural and geographic ties between trading partners matter more for differentiated than for homogeneous products, based on the classification proposed in [Rauch \(1999\)](#). A range of alternative models and specifications test the robustness of all these results. Finally, the significance of unobserved heterogeneity in product-level export flows points to the importance of using firm- as well as product-level data to understand export survival.

The article is organized as follows. Section I presents the dataset and provides descriptive statistics on the global pattern of duration dependence in the data as well as differences in the determinants across product groups. Section II briefly reviews the theoretical and the empirical literature on export market entry and exit. Section III discusses the econometric specification of a hazard model that incorporates country- and product-specific factors that influence export survival, and presents the empirical results, including a battery of robustness checks. Section IV outlines the need for additional research.

I. DATA AND PATTERNS OF EXPORT SURVIVAL

The empirical results of this article are based on a novel dataset that is more comprehensive in a number of important dimensions than those used in previous studies of trade flow survival. It includes data on export flows at the five-digit level of the Standard International Trade Classification (SITC) from 82 exporting countries to 53 importers. This amounts to a total of 2,861,394 export lifetime spells.²

The level of product detail in this dataset is finer than that of previous studies. It considers 1,271 products (petroleum products are excluded), whereas Blyde's (2008) dataset had 625 products and Besedes and Prusa's (2007) had 380.³ While it is interesting to see whether the conclusions of these previous studies hold for more detailed products, the more detailed product data are likely to better capture the dynamics of export entry and exit.

Coverage of exporters is also broader in this dataset than in previous studies, with the explicit intention of including countries with a wider range of income levels—in particular, including more Sub-Saharan African countries. [Blyde \(2008\)](#) has 47 high- and middle-income exporters and no low-income exporters, whereas [Besedes and Prusa \(2006b\)](#) investigate export survival

2. This study considers an export occurrence to be a flow greater than \$1,000, but it also tests the sensitivity of the conclusions to flow size.

3. [Besedes and Prusa \(2006b\)](#) and [Nitsch \(2007\)](#) use very detailed data at or close to the tariff-line level but analyze only the duration of exports to a single market, the United States and Germany, respectively.

patterns in 46 countries, only 1 of them (Madagascar) is low income. The dataset compiled for this study covers 82 exporters including 22 low-income countries and 22 Sub-Saharan African countries.

It is widely accepted that import data are more reliable than export data, especially for developing countries. Therefore, the dataset is based on mirror statistics from importing countries. It spans the 20 years from 1985 to 2005, sufficient to undertake meaningful survival analysis. From the United Nations Commodity Trade Statistics Database (Comtrade), all importers were included that reliably report annual data at the desired level of disaggregation over the entire sample period. This means that the number of importers in this dataset set is lower than in previous studies.⁴ This condition seeks to avoid the erroneous attribution of export entries and exits to a country that result from failing to report in a particular year.

Inevitably, some of the data are left or right censored. This is the case for about a quarter of the flows in the sample. Left censoring means that a flow is observed in the first year of the sample period but with no indication of when it came into existence. Right censoring implies that a flow is observed in the final year of the sample but with no indication of how long it will continue to exist. Right censoring is less of a problem since the duration estimation techniques use the information on the time of survival up to the censoring point but do not make any inference about what will happen to the export spell subsequently. Left censoring is a more serious problem because econometric techniques that deal efficiently with left-censored spells typically rely on strong additional assumptions or supplementary data, which are not available in this case.⁵ While left-censored observations are included in the descriptive statistics so as to present a complete picture of export survival rates, they are excluded from the sample for the econometric analysis. Strictly speaking, therefore, the

4. The need for consistent import data over 21 years means that most African countries and transition economies are absent from the importer group. The UN Comtrade database reveals that the largest sample of importers could be obtained for data reported according to the Standard International Trade Classification (SITC) as opposed to the Harmonised System, which was introduced more recently. Working with the SITC allows for the longest possible time dimension. As the SITC has been twice revised during the sample period, the data were combined using backwards concordance to Revision 1. In general, backwards concordance leads to more consistent product-level data over time but at the price of less detail. There is an issue regarding a small number of product categories that are joined together in subsequent revisions. There are around 100 categories where backwards concordance appears to be problematic, as shown by the appearance of zeros in the aggregate of imports from all countries in the sample. Including these product categories may therefore introduce some bias into the analysis of survival. On the other hand, the fact that these product categories are aggregated over time suggests that they have become less important and indeed may not contain important information. An extreme example is the product code for steam engines.

5. See, for example, D'Addio and Rosholm (2002).

conclusions about the determinants of export survival refer only to new exports rather than to all export flows in the sample.⁶ This strategy has been followed in much of the literature, although the distinction has not always been made clear.

Another analytical challenge concerns how to deal with one-year interruptions of export flows. Some researchers (Blyde 2008; Besedes and Prusa 2006b) have chosen to ignore interruptions of one year, since they may reflect simply reporting errors. This arbitrary strategy may create more problems than it solves. For the benefit of transparency, such gaps in the data are not ignored in analysis here, although the robustness of the results to their exclusion is examined.

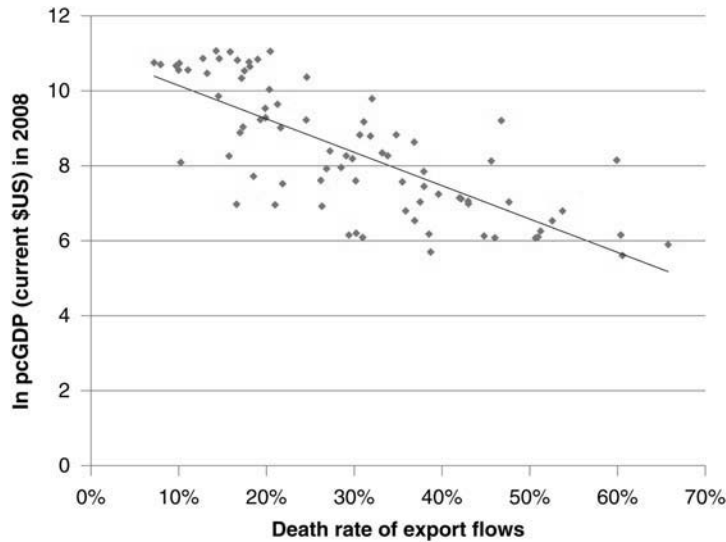
Plotting the average death rate of export flows over the sampling period for each country against per capita GDP (natural log) shows clearly that death rates are high across the sample (figure 1). The death rate is defined as the number of trade flows that cease to exist in a given year divided by the total number of trade flows and thus reflects the unconditional probability that an export flow disappears in that year. On average, about 15 percent of all export relationships in a given year disappear before the beginning of the next year. There is a strong negative correlation between death rates and per capita GDP. Death rates are below 10 percent in developed countries but as high as 50 percent and more in developing countries (in Mozambique and Zimbabwe, for example).

To gain a better understanding of the pattern of duration dependence in the data, a nonparametric estimate of the survivor function is investigated. The data on the duration of export flows are expressed in annual observations, indicating only that an export flow was recorded during the year. For flows that cease in a particular year, all that is known is their duration expressed in annual increments. For flows that survive, the available information is that their duration exceeded the lower limit of the last observed duration. In effect, the data can be interpreted as grouped survival time data, and so the life table method is appropriate to estimate the survival function nonparametrically. This method is based on the same idea as the Kaplan-Meier (1958) product-limit estimator for continuous time data but adjusts the estimate of the survivor function for the possibility that export flows could have ended at any point during the time interval. This boils down to using an average estimate centered on the midpoint of the interval.

The survivor function $S(t)$ is defined as the cumulative probability of surviving up to some point in time t , and T denotes the time to a failure event for a particular trade flow. Defining d_i as the number of failures observed in interval i , N_i as the number of spells at risk of failure at the start of the interval and n_i

6. In this study, the sample period of 20 years entails that some proportion of these “new exports” have existed for many years. Indeed, even datasets with longer sample periods will still be concerned only with “new exports.”

FIGURE 1. Relationship of Death Rate of Exports and per Capita GDP, 1985–2005



Source: World Bank various years and regression results based on data from UN Comtrade.

as the adjusted number of spells at risk of failure at the midpoint of the interval, one can write n_j as $n_j = N_j - \frac{d_j}{2}$ and the life table estimator of the survivor function as

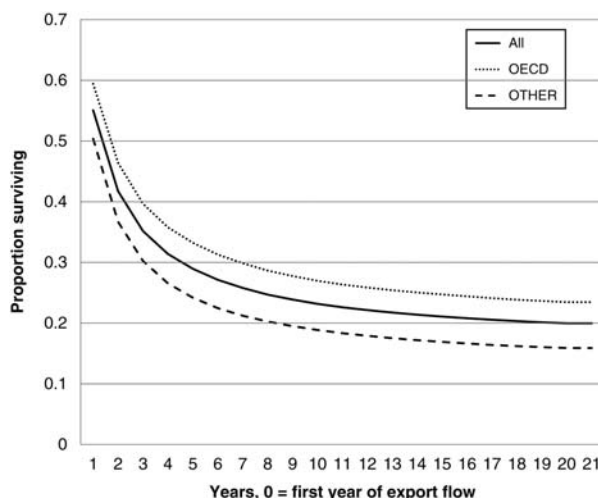
$$\hat{S}(t_j) = \prod_{i=1}^j \left(1 - \frac{d_i}{n_i}\right) \quad (1)$$

A plot of the life table survival function for the entire sample shows clear evidence of negative duration dependence throughout the lifetime of the trade flows (figure 2). In other words, the probability that an export flow will disappear falls the longer the duration of the flow. During the first five years of the emergence of a given trade flow, hazard rates are particularly high. Only about a third of all export spells survive this initial period; a fifth survive the first 20 years. This finding is in line with previous results in Besedes and Prusa (2006a) and Blyde (2008), among others.

Examination of the estimated rates of survival across income groups immediately reveals the presence of negative duration dependence (table 1). The estimates suggest that export flows are more likely to survive longer in richer countries. This finding holds for the entire sample and for each one-digit SITC sector.⁷ For high-income countries, 59 percent of trade flows

7. Similar patterns of survival rates within each sector suggest that the finding of lower survival rates for developing countries is not due simply to the types of products these countries tend to export.

FIGURE 2. Survival Rates of Export Flows, 1985–2005



Source: Regression results based on data from UN Comtrade.

survive longer than one year. For low-income countries, only 39 percent do. After 20 years, 23 percent of export flows survive for high income countries, but only 8 percent do for low-income countries. Survival rates for middle income countries fall in between, but closer to the upper end of the interval between high- and low-income countries. This could suggest that the problem of export survival is more relevant for countries moving from low- to middle-income status than for those moving from middle- to high-income status. A similar finding is that the survivor function for Organisation for Economic Co-operation and Development (OECD) countries lies consistently above the survivor function for the remaining countries (see figure 2).

Results are similar for individual countries, with 70 percent of export flow spells surviving for more than one year for the United States but less than 30 percent for Burkina Faso. Within 10 years of initiation, some 41 percent of export flows survive for the United States, but only 17 percent for Argentina, 12 percent for Egypt, and 4 percent for Burkina Faso.

The evidence in this section is suggestive. Section III investigates more deeply the factors that might shorten or lengthen the life of an export flow and explain differences in survival rates. The focus is on why export flows in low-income countries have substantially lower survival rates than those for middle- and high-income countries. But before this econometric analysis of the trade flow data, a review of the literature on the determinants of the duration of export flows sets the context.

TABLE 1. Export Survival Rates by One-Digit Standard International Trade Classification and Country Income Group, 1985–2005 (percent)

All goods										SITC 1			SITC 2			SITC 3			SITC 4		
Years	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income			
1	39	51	59	39	51	57	43	49	52	49	52	55	43	52	56						
5	15	25	32	15	24	31	15	21	24	23	24	27	18	24	29						
10	11	19	26	11	18	24	9	15	17	18	18	19	12	18	22						
15	9	18	24	8	16	21	7	13	14	16	16	16	11	15	19						
20	8	17	23	7	14	19	6	12	13	15	15	15	9	15	18						
21	8	17	23	7	14	19	6	12	13	15	15	15	9	15	18						
	SITC 5			SITC 6			SITC 7			SITC 8			SITC 9								
Years	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income	Low income	Middle income	High income						
1	36	52	59	39	51	58	34	50	62	40	52	60	46	52	58						
5	13	26	33	16	25	31	10	24	38	17	27	35	20	26	32						
10	9	20	26	12	19	25	7	20	33	13	22	29	14	21	25						
15	7	18	24	10	17	22	6	18	31	12	20	27	11	18	23						
20	6	17	23	9	16	21	5	18	30	11	19	26	10	17	22						
21	6	17	23	9	16	21	5	18	30	11	19	26	10	17	22						

Source: Authors' calculation based on data from UN Comtrade.

II. THE LITERATURE ON THE DETERMINANTS OF EXPORT SURVIVAL

International trade theory says little about the duration of trade flows. Recent developments in trade theory have focused on heterogeneous firms and entry into exporting and on how reductions in trade costs shift resources from lower productivity firms to higher productivity exporters. There has been little attention to why some firms cease exporting within a short time of starting to export. Similarly, the literature on export behavior, based extensively on firm surveys, has concentrated on differences between exporters and nonexporters and on whether the observed higher productivity of exporters is a cause or result of entry into foreign markets. Much less information is available on the evolution of exporters after they enter into trade activities, reflecting in part the lack of continuous surveys.

Besedes and Prusa (2006b) were the first to investigate the duration of U.S. import trade, showing that short survival times are consistent with a search model of international trade. In particular, the high number of short trade relationships appears to reflect uncertainty in the formation of trade relationships. Both higher reliability and lower search costs lead to larger initial purchases and longer relationships. This finding is confirmed in follow-up studies such as Blyde (2008) and is investigated in detail by Besedes (2008).

Market knowledge and other information needed for successful entry into exporting are likely to be important in explaining exit. With less than perfect information about the fixed costs of exporting a particular product to a particular market or some uncertainty about the value of these costs, lower productivity firms that are marginal entrants into exporting may not survive. Indeed, firms lacking full market information may use entry into export markets as cost-discovery mechanism for exporting to a particular market and withdraw if they find that incurring the fixed costs of exporting to that market is unprofitable. Initial entry is likely to be small scale and exit prevalent. Short-term entry may also reflect the search processes that are necessary to match suppliers and buyers in overseas markets.

When information on the costs of exporting is readily available or obtainable at little cost, larger scale entry is more likely and rapid exit should be less frequent. The greater the export activity in a particular overseas market—the more exporters of other products to that market and the greater the overall experience with exporting a particular product—the easier it should be to obtain such information. Several recent studies have sought to formalize the influence of imperfect information on the dynamics of export entry and exit.

Rauch and Watson (2003) look at the initiation of export flows from the perspective of developed country buyers that are uncertain about the long-term prospects of a potential partnership with developing country suppliers. The buyer is unsure whether the supplier will be able to deliver large orders to the buyer's specifications even if the buyer invests in training the supplier. The buyer may glean information about the capacity of the supplier before making

such an investment by starting with small orders that generate no profits but that reveal whether the training is likely to be successful. In other words, the buyer has the choice of starting big or small. Finally, the buyer has the option of continuing a supplier relationship or abandoning it and searching for a new supplier. Once a successful relationship is in place, the buyer can access a network of other suppliers and can obtain information on new firms without incurring search costs.

The model suggests that buyers in importing countries are more likely to start a relationship with an exporter with small orders the higher the search cost and the lower the probability that the supplier will be able to meet the buyer's requirements. But the model also predicts that export flows that commence with large orders will last longer—because buyers tend to initiate large orders with suppliers with lower production costs and will be less likely to look for an alternative supplier.

[Araujo and Ornelas \(2007\)](#) characterize information costs relating to exporting in terms of weak contract enforcement. Potential exporters look for partnerships with buyers in overseas markets, but the weak institutional environment allows some buyers with little concern for the future to behave opportunistically and to default. In such a climate, forward-looking buyers seek to differentiate themselves from myopic buyers by building a reputation over time. Hence, information costs decline as exporting experience accumulates. Initially small export flows increase over time as the exporter learns more about the buyer's trustworthiness and probability of defaulting on the contract. Thus, the longer the partnership with the buyer continues, the lower the probability of exiting from exporting.

Improving the institutions for contract enforcement has a direct, positive effect on exports by reducing uncertainty and raising the expected return of the exporter. However, there is also an indirect negative impact on the size of exports since stronger contract enforcement reduces the value of future reputation compared with the situation under weaker enforcement. In other words, it becomes more difficult for a buyer that has built a reputation to differentiate itself from other buyers that are now less likely to default because of the threat of a more effective legal challenge. The latter effect will tend to be stronger in cases of initially weak enforcement. Stricter enforcement of contracts also increases the level of initial exports in a new partnership and supports relationships of longer duration.

In a recent study, [Albornoz, Pardo, and Corcos \(2010\)](#) suggest that firms discover their profitability as exporters only after engaging in exporting. Firms adjust quantities and decide whether to exit or to serve new destinations based on what they learn about themselves through exporting experience. [Edwards \(2007\)](#) discusses the impacts of trade liberalization in a situation of search costs and matching of producers and buyers in overseas markets. Once a search process has been under way for a long time, a proportion of firms and buyers will be matched and so will be less sensitive to new potential trade

partners. This would explain the negative duration dependence found by Besedes and Prusa (2006b); once a flow has been established for a while, the probability of failure drops. Firms in established relationships are likely to be less sensitive to changes in the relative prices of different suppliers than are firms and distributors that are still looking for a satisfactory partner. This suggests time dependency and that inappropriate sequencing of trade policy reforms can have adverse long-run impacts. For example, preferential import liberalization may lock in trade diversion (due to information costs) even after subsequent multilateral tariff reductions have removed tariff preferences.

Information asymmetries and formal institutions for contract enforcement tend to be more important for buyers searching among suppliers providing products of different quality. Put differently, matching suppliers and buyers will tend to be easier for standard and homogeneous products. Much of the discussion on export diversification in developing countries centers on increasing exports of differentiated products, especially manufactures, and reducing the importance of homogeneous products, particularly commodities. Rauch (1999) presents evidence, albeit tentative, that search costs are higher and matching is more difficult for differentiated products and that proximity, common language, and colonial links are more important for differentiated products than for homogeneous products that are traded on organized international exchanges.

Quality introduces another set of institutions that might influence bilateral trade. Poor quality metrology, testing, and conformity assessment facilities in developing countries mean that exporters will incur additional costs for assessing quality and conformity with private or public standards when exporting products to more developed countries or that some uncertainty will surround issues of quality and standards. The buyer may have additional uncertainty about the ability of the exporter to consistently deliver the specified product quality. These concerns about information and quality are likely to be most important for trade between countries with different quality preferences and different stages of development of the institutions responsible for contract enforcement and quality.

The discussion above and the studies by Besedes and Prusa (2006b), Blyde (2008), and Brenton, Pierola, and von Uexkull (2009) suggest that the following variables likely influence the hazard rate of export flows:⁸

- The initial value of the trade relationship is a crude proxy for the level of confidence the trading partners originally had in the profitability of the trade relationship.
- The distance between the exporter and the importer captures two types of costs that a firm might encounter in a trade relationship: trade costs and

8. Precise definitions of these variables are provided in appendix 1. The reference year for all variables is the year in which the trade relationship starts or, in the case of left-censored observations, 1985, the beginning of the sampling period.

the initial cost of searching for an appropriate trading partner. These costs are likely to be lower for neighboring countries that share a common language or common border and countries with past colonial ties.

- For both trading partners, larger economic size and greater wealth should facilitate the survival of trade flows. Bigger markets are likely to have a larger number of buyers, thus increasing the likelihood that the exporter will find a suitable match. This effect is captured by the inclusion in the model of the trading partners' GDP (this effect is analyzed more closely through robustness tests).
- Exporting experience in general and at the product and industry level in particular is captured by the amount of total exports to the market and total exports of the product (to all markets). Other important determinants are the size of the import market and the exporter's knowledge of the market. To account for these factors, the total value of trade between the trading partners and the global value of the importer's imports of the product concerned are included. The larger these flows are, the greater the potential spillovers of information to new entrants.
- If market experience with a product is specific to the importing region, existing export relationships with countries that border the import market should facilitate survival in a new market. A test is conducted for whether exporting experience follows a geographic pattern. A dummy variable indicates whether the exporter is already exporting the product concerned to a neighboring country of the importing country.
- Trial and error patterns of learning to export to a market are captured by a dummy variable indicating whether there have been previous exporting spells of the product to the importer.
- Several policy variables are included: a measure of the variability of the bilateral exchange rate when the flow is initiated; a measure of the deviation of the bilateral exchange rate from the period average in the year of entry into the exporting relationship, as a crude proxy for exchange rate misalignment; and a dummy variable for the presence of a preferential trade agreement between the trading partners, as an indicator of trade policy restrictiveness. While it would be preferable to include the actual tariff rate in the import market for each product, this information is available only for a subset of countries and years. Including this information would come at the expense of a greatly reduced sample size.

III. ECONOMETRIC SPECIFICATION AND EMPIRICAL RESULTS

Previous studies of export survival, including those by [Besedes and Prusa \(2006a, b\)](#) and [Blyde \(2008\)](#), have used the continuous time proportional hazards model proposed by [Cox \(1972\)](#), which is widely used in survival analysis. This model is based on the assumption that variables influencing

survival have a proportionate impact on the base hazard function; that is, a change in variable z would raise or lower the hazard function by the same factor in any period. The mathematical form of the hazard function for export flow i in the Cox model is

$$\lambda_i(t) = \lambda_0(t)\exp(Z_i(t)'\beta) \quad (2)$$

where $\lambda_0(t)$ is the baseline hazard at time t , $z_i(t)$ is the vector of covariates for flow i that have a proportional impact on the hazard function, and β is a vector of coefficients to be estimated that characterizes how z affects the hazard function.

The Cox model is widely used because it enables estimating the relationship between the hazard rate and a set of explanatory variables using a partial likelihood approach without having to assume anything about the shape of the baseline hazard function. This is why the model is often referred to as a semi-parametric model. However, this convenience relies heavily on the assumption that the baseline hazard function summarizing the pattern of duration dependence can be separated from the individual specific non-negative function of covariates. Hence, the function of covariates scales the baseline survivor function with a constant factor independent of survival time. If the data are not consistent with this assumption, the model is misspecified. Possible reasons for the failure of this assumption could be the omission of relevant time-dependent variables or the presence of unobserved individual heterogeneity. Previous studies of export flow survival at the product level have not tested the proportional hazards assumption and have not investigated unobserved individual heterogeneity.

The first step is to estimate a simple Cox model. The sample is stratified by the 1,271 product categories. Hence, the estimation does not insist that the baseline hazard be proportional across products but allows a separate baseline hazard function for each product group. The number of observations in the dataset makes it possible to use such fine stratification without incurring a problematic drop in degrees of freedom.

The Cox estimation results are presented in the first set of columns in table 2 for the preferred specification. The coefficient estimates are not discussed at this point as the purpose of this estimation is to test for the validity of the model specification—the validity of the proportional hazards assumption. A comprehensive discussion of applicable testing procedures for the Cox model can be found in [Horowitz and Neumann \(1992\)](#). An interesting first descriptive check is a plot of the log of nonparametric estimates of the hazard function against time for different subgroups of trade flows in the sample. If the proportional hazards assumption is appropriate, the plots should move in parallel. Although not reported here, different subgroups of the sample were analyzed, and the evidence from these plots does not support the proportional hazards assumption.

TABLE 2. Estimation Results, 1985–2005

Variable	1. Cox (1972) proportional hazards model			2. Prentice-Gloekner model without unobserved heterogeneity ^a			3. Prentice-Gloekner model with unobserved heterogeneity, gamma distribution ^a			4. Prentice-Gloekner model with unobserved heterogeneity, gamma distribution ^a			5. Multiple spells excluded			6. Small trade flows excluded (<\$55,000)		
	Exp(b)	p-value	25th–>75 th percentile ^b (%)	Exp(b)	p-value	25th–>75 th percentile ^b (%)	Exp(b)	p-value	25th–>75 th percentile ^b (%)	Exp(b)	p-value	25th–>75 th percentile ^b (%)	Exp(b)	p-value	25th–>75 th percentile ^b (%)	Exp(b)	p-value	25th–>75 th percentile ^b (%)
Initial value	0.908	0.000	–22.5	0.966	0.000	–8.3	0.968	0.000	–7.7	0.967	0.000	–8.0	0.987	0.000	–3.1	1.155	0.000	37.9
Distance	1.074	0.000	10.2	1.191	0.000	26.3	1.254	0.000	35.0	1.224	0.000	30.8	1.310	0.000	42.7	1.280	0.000	38.5
Border	0.831	0.000	–16.9	0.670	0.000	–33.0	0.595	0.000	–40.5	0.627	0.000	–37.3	0.544	0.000	–45.6	0.556	0.000	–44.4
Common language	0.895	0.000	–10.5	0.739	0.000	–26.1	0.672	0.000	–32.8	0.701	0.000	–29.9	0.637	0.000	–36.3	0.679	0.000	–32.1
Colony	0.988	0.001	–1.2	0.976	0.000	–2.4	0.976	0.000	–2.4	0.977	0.000	–2.3	0.975	0.040	–2.5	1.073	0.000	7.3
GDP product	0.979	0.000	–5.6	0.946	0.000	–14.3	0.928	0.000	–19.1	0.936	0.000	–16.9	0.909	0.000	–24.2	0.929	0.000	–18.7
Total exports	0.989	0.000	0.0	0.974	0.000	0.0	0.968	0.000	0.0	0.971	0.000	–14.7	0.964	0.000	–17.8	0.965	0.000	–17.4
Industry exports	0.997	0.000	–1.4	0.981	0.000	–7.4	0.965	0.000	–14.0	0.972	0.000	–11.3	0.956	0.000	–17.8	0.952	0.000	–19.4
Product exports	0.942	0.000	–22.8	0.863	0.000	–54.3	0.835	0.000	–65.3	0.848	0.000	–60.1	0.816	0.000	–72.9	0.811	0.000	–75.0
Product imports	0.984	0.000	–5.7	0.929	0.000	–26.0	0.905	0.000	–34.9	0.915	0.000	–31.1	0.886	0.000	–41.6	0.907	0.000	–34.0
Bilateral trade	0.965	0.000	–10.6	0.958	0.000	–12.7	0.946	0.000	–16.4	0.951	0.000	–14.7	0.947	0.000	–16.2	0.936	0.000	–19.3
Neighbor	0.881	0.000	–11.9	0.729	0.000	–27.1	0.682	0.000	–31.8	0.703	0.000	–29.7	0.600	0.000	–40.0	0.628	0.000	–37.2
Previous spell	0.842	0.000	–15.8	0.351	0.000	–64.9	0.274	0.000	–72.6	0.305	0.000	–69.5	0.315	0.000	–68.5	0.315	0.000	–68.5
FX volatility	1.007	0.000	0.5	0.992	0.000	–0.6	0.990	0.000	–0.8	0.990	0.000	–0.8	0.997	0.169	–0.3	0.991	0.006	–0.7
FX misalignment	1.000	0.160	0.0	0.996	0.000	–0.4	0.997	0.000	–0.3	0.997	0.000	–0.3	0.998	0.003	–0.2	0.992	0.000	–0.8
PTA	0.999	0.599	–0.1	0.961	0.000	–3.9	0.955	0.000	–4.5	0.957	0.000	–4.3	0.912	0.000	–8.8	0.959	0.000	–4.1
logtime				0.893	0.000		1.186	0.000										
No. of observations	2,861,394			2,861,394 ^c			2,861,394 ^c			2,861,394 ^c		2,702,156			1,762,595			
LogL				–2,133,032			–2,121,082			–2,123,406		–968,098			–400,017			
gamma variance																		
Exp(b)							0.471			0.291		1.364			0.795			
p-value (Exp (b))							0.000			0.000		0.000			0.000			
LR test																		
chi ²							23899			25197		116,813			11,205			
p-value (chi ²)							0.000			0.000		0.000			0.000			

Note: See appendix 1 for variable sources and definitions.

a. The Prentice-Gloekner regressions include a set of biannual calendar time dummy variables (not reported).

b. Percentage change in the hazard ratio if the value of the exogenous variables increases from the 25th percentile to the 75th percentile of all observations in the dataset. In the case of dummy variables, the impact of a change from 0 to 1 is reported.

c. For the Prentice-Gloekner estimation, the dataset is modified from one observation per export spell to one observation per year with export spell, which increases the number of observations to 7,234,950.

Source: Regression results based on data from UN Comtrade.

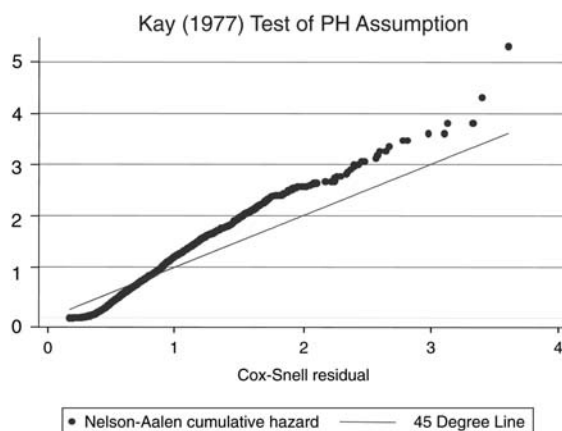
Kay (1977) suggests another testing procedure. He derives residuals for the Cox model and tests the proportional hazards assumption by plotting these so-called Cox-Snell (1968) residuals against the Nelson-Aalen cumulative hazard function (Nelson 1972; Aalen 1978). For the proportional hazards assumption to be valid, the plot should deviate little from the 45 degree line. This condition is clearly violated in the data (figure 3). This result holds independently of whether the sample is stratified. Different combinations of covariates, including interaction terms and time-dependent variables, were used to capture as much information as possible, and all led to the same conclusion.

A formal statistical testing procedure of the proportional hazards assumption in the Cox model is based on residuals derived by Schoenfeld (1982) that do not vary over time if the model specification is correct. Table 3 presents the results of testing for a zero slope of the plot of the Schoenfeld residuals against survival time both globally and for each covariate. The test statistics and the associated *p*-values reveal that the null hypothesis of a constant slope is rejected with confidence for the model as a whole and for all but two covariates individually.

These results constitute strong evidence for the failure of the proportional hazards assumption in the data. In the above testing procedures, both time-dependent covariates and interaction terms appeared to be neither the reason for this problem nor its solution. Another possible cause of the failure of the proportional hazards assumption is the presence of unobserved individual heterogeneity—which could arise, for example, from omitted firm-specific variables, such as managerial capacity—that captures the risk of failure in addition to that captured by the observed covariates.

In general, failure to account for unobserved heterogeneity will, if it is indeed important, lead to overestimating the degree of negative duration

FIGURE 3. Kay (1977) Test of Proportional Hazards Assumption, 1985–2005



Source: Regression results based on data from UN Comtrade.

TABLE 3. Schoenfeld (1982) Test of Proportional Hazard Assumption, 1985–2005

Variable	Chi ²	p-value
Initial value	513.75	0.000
Distance	691.09	0.000
Border	248.71	0.000
Common language	149.33	0.000
Colony	1.41	0.200
GDP product	4,451.92	0.000
Total exports	4,634.23	0.000
Industry exports	167.37	0.000
Product exports	628.62	0.000
Product imports	240.02	0.000
Bilateral trade	266.56	0.000
Neighbor	124.87	0.000
Previous spell	436.64	0.000
FX volatility	0.000	0.900
FX misalignment	865.75	0.000
PTA	11.54	0.000
Global	14,551.38	0.000

Source: Regression results based on data from UN Comtrade.

dependence in the (true) baseline hazard and underestimating the degree of positive duration dependence. In other words, the coefficient on the functional representation of analytical time will be biased downward for export flow data that are subject to negative duration dependence in the hazard. Moreover, in the presence of unobserved and unaccounted for individual heterogeneity, the proportionate effect of a given regressor on the hazard rate is no longer constant and independent of survival time, and the proportionate response of the hazard to variation in each regressor at any survival time is attenuated.

The next step is to estimate a model that addresses unobserved heterogeneity. A discrete time model is more appropriate for trade flow data than the continuous time model applied by Besedes and Prusa (2006b) and followed by others such as Blyde (2008). As discussed above, the data are organized in annual observations. There are also many tied failure times. Such data are properly accommodated in the framework of discrete duration models (see Lancaster 1990). For this reason, an appropriate starting point for the estimations is the Prentice-Gloekler (1978) model,⁹ which is a discrete time equivalent of the continuous time proportional hazards model. If unobserved heterogeneity is assumed to take a multiplicative form, the hazard function can

9. The Prentice-Gloekler model is an interesting complement to the Cox model, as both estimators make no assumption about the shape of the baseline hazard. For suitably re-organized data in export flow–period format, the Prentice-Gloekler log-likelihood function is the same as the log-likelihood function for a generalized linear model of the binomial family with a complementary log-log link (Jenkins 1995).

be expressed as

$$\lambda_i(t) = \theta_i \lambda_0(t) \exp(Z_i(t)' \beta) \quad (3)$$

where θ_i is an unobserved random variable that is assumed to be independent of $z_i(t)$, the vector of covariates for flow i . Maximum likelihood estimates of the parameter vector and baseline hazard can be obtained by conditioning the likelihood function on θ_i and then integrating over the distribution of θ . This approach requires specifying a distribution function for θ . Following Meyer (1990), a common approach is to use the gamma distribution with mean 1 and variance σ^2 , which gives a closed form expression for the likelihood function.

Table 2 also presents the results from estimating the preferred specification using both the Prentice-Gloeckler model and the Prentice-Gloeckler model incorporating a gamma mixture distribution summarizing individual heterogeneity.¹⁰ The log of time was introduced as an additional variable into one specification to directly capture the pattern of duration dependence (second and third set of columns in table 2), although the benchmark specification (discussed below) does not contain this variable. As can be observed, the results are not very sensitive to the exclusion or inclusion of this additional variable. Finally, biannual calendar time dummy variables are also included in all regressions with the Prentice-Gloeckner model to account for the possibility that the survival of export flows is sensitive not only to analytical time but also, for instance, to cyclical downturns in global demand.

A first glance at the results reveals two key findings. First, the p -values for all variables are virtually zero and the likelihood ratio test strongly rejects the null hypothesis of the absence of unobserved individual heterogeneity in the data. This result is confirmed when a normal distribution is used instead of the gamma mixture distribution to summarize unobserved heterogeneity (not reported). Second, the proportional response of the hazard to changes in almost all of the regressors is underestimated in the basic Prentice-Gloeckler model. Furthermore, the degree of negative duration dependence in the hazard is overestimated when unobserved individual heterogeneity is not taken into account. In fact, the models indicate that the strongly negative duration dependence found in the survivor functions in the previous section is well explained by the regressors once unobserved heterogeneity is taken into account. The coefficient on the log of time suggests a slightly upward sloping baseline hazard.

The heterogeneity terms are significant when examining different product or income groups and when excluding very small trade flows or trade flows with previous spells in the sample (last two sets of columns in table 2). Learning more about the source of the heterogeneity is thus an important avenue for future research. It is likely that it can be found in the failure to account for

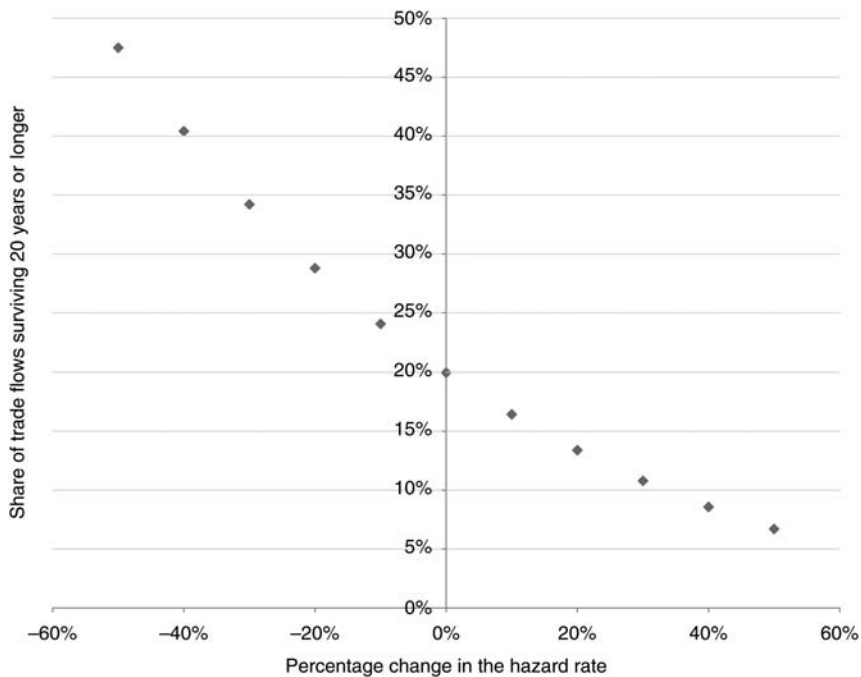
10. Throughout this article, the estimated coefficients are presented in exponential form.

firm-specific characteristics in product-level data. It is interesting that recent application of a discrete time proportional hazards model to firm-level export data for Spain (Esteve-Perez, Pallardo, and Requena 2008) indeed found no evidence of unobserved heterogeneity.

The remainder of this section discusses the results of the preferred specification, the Prentice-Gloeckler model using a gamma mixture distribution to summarize unobserved individual heterogeneity (fourth set of columns in table 2). To help interpret the economic significance of the results, the predicted percentage change in the hazard rate resulting from a move from the 25th percentile to the 75th percentile of the distribution of a given control variable (or from 0 to 1 in the case of dummy variables) is shown in table 2. It shows, for example, that the hazard rate is 31 percent higher for a trade relationship with a country in the 75th percentile of the distance variable than with a country in the 25th percentile.

Figure 4 links changes in the hazard rate to the share of export flows that survive for 20 years or more assuming a hazard rate identical to the one for the sample as a whole. It shows that the number of export relationships surviving after 20 years falls from about 20 percent to around 12 percent with a 31 percent increase in the overall hazard rate for the dataset. The impact of a

FIGURE 4. Impact of a Change in the Hazard Rate on Survival Time of Trade Flows



Source: Authors' calculation.

given percentage change in the hazard rate on a country's overall export performance will of course depend on several additional factors, such as the country's individual hazard rate, the number of existing and new trade relationships in each year, and the assumed growth rate of export flows that survive.

The results support the prediction that hazard rates are lower for export flows with larger initial values. Moving from the 25th to the 75th percentile for initial export values reduces the hazard rate by 8 percent. However, this effect is much weaker than the one predicted by the Cox model (–23 percent).¹¹ The hazard rate is higher for initially small flows, and this result holds until all trade flows with initial values smaller than \$55,000 are excluded from the sample (sixth set of columns in table 2). The hazard rate also increases strongly with distance. Intuitively, trade costs would be expected to be the main driver of this effect. A common language (–30 percent) and a common border (–37 percent) significantly reduce the hazard rate, whereas colonial ties have only a small impact (–2 percent). Qualitatively, all these results are consistent with those found by Besedes and Prusa (2006a) and Blyde (2008). However, the coefficient estimates differ markedly in some cases.

The results also support the hypothesis that previous export experience is important in maintaining a trade relationship. Having previous export spells in the same product category decreases the hazard rate by 70 percent. Moreover, the results suggest that exporting experience is product rather than market specific. A move from the 25th to the 75th percentile for a country's total exports of goods within the same five-digit product category translates into a 60 percent reduction in the hazard rate. The impact of industry-specific and global export experience is weaker but not negligible. The size of the importer's market for a given product also appears to matter. Finally, these results also support the assumption that the economic size of both trading partners contributes to trade flow facilitation and to trade flow survival (see Blyde 2008 and Baldwin and Harrigan 2007, for example). A likely reason is that the number of possible matches between firms increases with the size of the two economies.

The analysis also finds that the hazard rate is reduced (by 30 percent) if the exporter already sells the product to other countries in the same region as the importer. This suggests that the regional nature of exporting experience should not be neglected. There appear to be learning effects specific to the product and the importing regions that help exporters sustain their trade relationships. This complements the finding of Roberts and Tybout (1995), who show that experience matters for initiating trade flows as well.

The results also provide evidence that both an initial overvaluation of the exporter's exchange rate and higher exchange rate volatility during initiation of

11. The importance of the initial size of exports is consistent with work by Brenton and von Uexkull (2009), who find that technical assistance targeted to specific export products tends to be more effective for the subsequent growth of exports for larger export flows.

export flows decrease hazard rates, but these effects do not seem to be economically significant. Finally, the coefficient on the preferential trade agreement dummy variable (*PTA*) indicates that a preferential trade agreement between the exporter and the importer reduces hazard rates by about 4 percent. This effect seems weaker than one would expect *ex ante*. In fact, the true effect may be attenuated by the definition of the variable, which gives the reference year for the *PTA* dummy variable as the starting year of the trade relationship. This implies that trade flows that become subject to a trade agreement after they are initiated are recorded as not being subject to the agreement.¹²

Two robustness checks were performed for the preferred specification (last two sets of columns in table 2). The first test excludes multiple spells from the analysis; doing so does not have a strong effect on the results. The second test excludes initially small trade flows (less than \$55,000), which reverses the results for the impact of the initial value. This suggests that the positive impact of the initial export value is driven by a large number of observations with very small initial values and short duration but that once the initial size exceeds a certain threshold, the impact is reversed. A large share of these very small trade flows may be trial and error attempts at exporting, with firms exporting small amounts (or buyers ordering small amounts) initially to test whether a trade relationship is viable. This effect is no longer relevant in the data once these very small transactions are excluded. To the contrary, some trade flows with the largest initial values in the sample also have very short duration, which explains why the coefficient on the initial value is reversed once small initial value trade flows are dropped.

Previous research has argued convincingly that factors such as common language, common border, proximity, and other indicators of search barriers to trade may play a greater role for differentiated goods than for homogeneous goods (see Besedes and Prusa 2006a). This notion is investigated formally by dividing the sample into three sub-samples, each corresponding to one of the product categories classified in Rauch (1999).¹³ The differences across product categories are small but generally confirm that factors such as distance, common border, common language, and colonial ties that influence trade costs play a more decisive role in the survival pattern of differentiated products than of homogeneous products (table 4). Intuitively, differentiated products are not traded on organized exchanges, so the exporter has to gather information and engage in an extensive search process, both in initiating trade flows and in sustaining them. Factors facilitating this search then contribute to a higher probability of trade flow survival.

12. This is particularly problematic for trade relationships that last for the entire sampling period. Many have a 0 for the *PTA* dummy because the number of preferential trade agreements in 1985 was much lower than it is today. As an illustration, only about 16 percent of trade relationships in 1985 were under a preferential trade agreement, whereas 40 percent were in 2005.

13. Experiments with both of Rauch's classification schemes shows that the results are not sensitive to the classification scheme chosen. Results are reported for the conservative classification.

TABLE 4. Results by Product Category at Preferred Specification (Prentice-Gloekner model with unobserved heterogeneity, gamma distribution), 1985–2005

Variable	Rauch: diversified				Rauch: reference priced				Rauch: homogeneous			
	95%		25th–>75th percentile ^a		95%		25th–>75th percentile ^a		95%		25th–>75th percentile ^a	
	Exp(b)	confidence interval	p-value	(%)	Exp(b)	confidence interval	p-value	(%)	Exp(b)	confidence interval	p-value	(%)
Initial value	0.943	0.941	0.945	0.000	0.934	0.931	0.937	0.000	0.969	0.963	0.975	0.000
Distance	1.241	1.235	1.248	0.000	1.244	1.233	1.254	0.000	1.219	1.197	1.241	0.000
Border	0.621	0.609	0.633	0.000	0.632	0.615	0.650	0.000	0.635	0.601	0.671	0.000
Common language	0.698	0.691	0.705	0.000	0.700	0.689	0.711	0.000	0.749	0.724	0.775	0.000
Colony	0.950	0.934	0.965	0.000	1.029	1.004	1.054	0.023	1.029	0.978	1.083	0.271
GDP Imp x Exp	0.921	0.918	0.923	0.000	0.945	0.941	0.950	0.000	0.954	0.944	0.964	0.000
Total exports	0.969	0.968	0.969	0.000	0.977	0.976	0.978	0.000	0.987	0.985	0.989	0.000
Industry exports	0.989	0.988	0.990	0.000	0.966	0.964	0.968	0.000	0.989	0.984	0.994	0.000
Product exports	0.828	0.827	0.829	0.000	0.866	0.864	0.868	0.000	0.883	0.878	0.887	0.000
Product imports	0.916	0.915	0.918	0.000	0.936	0.934	0.938	0.000	0.960	0.955	0.964	0.000
Bilateral trade	0.940	0.938	0.942	0.000	0.945	0.942	0.949	0.000	0.936	0.927	0.944	0.000
Neighbor	0.730	0.725	0.736	0.000	0.711	0.703	0.720	0.000	0.706	0.687	0.725	0.000
Previous spell	0.315	0.312	0.317	0.000	0.315	0.312	0.319	0.000	0.291	0.284	0.298	0.000
FX volatility	0.990	0.987	0.993	0.000	0.994	0.989	0.998	0.008	1.013	1.003	1.023	0.010
FX misalignment	0.995	0.994	0.996	0.000	0.998	0.996	0.999	0.002	0.990	0.987	0.993	0.000
PTA	0.968	0.960	0.976	0.000	0.957	0.945	0.970	0.000	0.996	0.967	1.025	0.763
No. of observations	4,745,333				1,707,599				347,020			
LogL	–1,302,371				–543,221				–121,643			
Exp(b)	0.230				0.320				0.450			
p-value (Exp(b))	0.000				0.000				0.000			
chi ²	1,001				8,734				3,928			
p-value (chi ²)	0.000				0.000				0.000			

Note: All regressions include a set of biannual calendar time dummy variables (not reported).

a. Percentage change in the hazard ratio if the value of the exogenous variables increases from the 25th percentile to the 75th percentile of all observations in the dataset. In the case of dummy variables, the impact of a change from 0 to 1 is reported.

Source: Regression results based on data from UN Comtrade.

It also appears that the importance of the initial trade value to the length of survival is lower for homogeneous products, which may be due to less need for a trial and error search process for a commodity traded on an organized international exchange. Similar arguments can be made for the variables characterizing the exporting experience at the product level, economic size of both trading partners, and the size of the import market for the given product. The negative effect of the *PTA* dummy variable on the hazard rate also becomes markedly weaker for homogeneous goods, perhaps because a preferential trade agreement might be less important in facilitating trade for homogeneous goods typically traded on organized exchanges.

Finally, the analysis considers exporters from different income groups (table 5). The goal here is to understand whether the factors affecting export survival influence exports from developing countries more strongly than exports from richer countries. The results suggest that both regional exporting experience and previous trade relationships between two trading partners are more important for low-income exporters. The differences in point estimates are substantial, as evidenced by the absence of any overlap of confidence intervals between the estimates for high- and low-income exporters. Similarly, it is perhaps not surprising to find that past colonial ties help high- and middle-income exporters much less than low-income exporters in sustaining their export flows; again, the confidence intervals do not overlap. A common border between trading partners has a significantly smaller positive impact on the probability that a trade flow will survive when the exporter is a developing country. This suggests that low-income countries still have a long way to go to fully exploit the advantages of lower trade costs when trading with neighboring countries.

Although perhaps not economically significant, the size of the initial trade flow has a positive effect on the hazard rate for low-income exporters. In non-reported regressions, this issue was further investigated by interacting the initial export value with the product category according to Rauch (1999). As it turns out, the slightly positive effect is driven only by homogeneous products that are traded on organized exchanges and that constitute a high share of developing country exports. One explanation could be that the recorded sources and destinations of these exports change frequently as buyers regularly switch purchases between different source countries.

Furthermore, the relationship between the *PTA* dummy variable and the hazard rate changes sign when the sample is restricted to low-income exporters. This finding is somewhat surprising. Although not reported here, this was analyzed further by interacting the *PTA* dummy variable with characteristics of the importing country. It appears that the effect of a preferential trade agreement on the hazard function is only positive for preferential trade agreements between low-income exporters and low-income importers. It could be that a preferential trade agreement between low-income countries encourages exports to more hazardous partner markets.

TABLE 5. Results by Exporter Country Category at Preferred Specification (Prentice-Gloekner with unobserved heterogeneity, gamma distribution), 1985–2005

Variable	High-income exporters				Middle-income exporters				Low-income exporters						
	Exp(b)	95% confidence interval		25th–> 75th ^a (%)	p-value	Exp(b)	95% confidence interval		25th–> 75th ^a (%)	p-value	Exp(b)	95% confidence interval		25th–> 75th ^a (%)	
Initial value	0.961	0.958	0.962	0.000	–9.6	0.974	0.971	0.977	0.000	–6.3	1.021	1.014	1.028	0.000	5
Distance	1.141	1.134	1.147	0.000	19.4	1.331	1.321	1.340	0.000	45.6	1.264	1.238	1.291	0.000	36
Border	0.663	0.645	0.682	0.000	–33.7	0.715	0.701	0.729	0.000	–28.5	0.766	0.703	0.834	0.000	–23
Common language	0.730	0.721	0.739	0.000	–27.0	0.761	0.751	0.771	0.000	–23.9	0.859	0.835	0.884	0.000	–14
Colony	0.938	0.922	0.954	0.000	–6.2	0.996	0.973	1.020	0.758	–0.4	0.777	0.746	0.809	0.000	–22
GDP product	0.959	0.956	0.962	0.000	–10.8	0.941	0.937	0.944	0.000	–15.6	0.915	0.908	0.923	0.000	–22
Total exports	0.976	0.974	0.977	0.000	–12.2	0.974	0.973	0.975	0.000	–12.9	0.980	0.979	0.981	0.000	–10
Industry exports	0.972	0.971	0.974	0.000	–11.1	0.958	0.956	0.960	0.000	–16.8	0.962	0.959	0.966	0.000	–15
Product exports	0.845	0.843	0.846	0.000	–61.6	0.840	0.838	0.842	0.000	–63.5	0.839	0.834	0.843	0.000	–64
Product imports	0.920	0.919	0.922	0.000	–29.2	0.919	0.917	0.921	0.000	–29.8	0.935	0.931	0.939	0.000	–24
Bilateral trade	0.914	0.911	0.917	0.000	–25.9	0.938	0.935	0.941	0.000	–18.7	0.978	0.973	0.984	0.000	–7
Neighbor	0.752	0.746	0.757	0.000	–24.8	0.658	0.651	0.664	0.000	–34.2	0.629	0.611	0.648	0.000	–37
Previous spell	0.321	0.319	0.323	0.000	–67.9	0.297	0.295	0.300	0.000	–70.3	0.270	0.264	0.276	0.000	–73
FX volatility	0.975	0.972	0.979	0.000	–1.9	1.039	1.034	1.043	0.000	3.0	1.008	1.002	1.014	0.014	1
FX misalignment	1.003	1.002	1.004	0.000	0.3	0.990	0.989	0.992	0.000	–0.9	0.977	0.973	0.982	0.000	–2
PTA	0.893	0.885	0.902	0.000	–10.7	0.976	0.966	0.986	0.000	–2.4	1.155	1.126	1.184	0.000	15
No. of observations	4,094,474					2,596,781					317,203				
LogL	–1,162,169					–764,662					–130,293				
Exp(b)	0.156					0.400					0.410				
p-value	0.000					0.000					0.000				
chi2	3,895					1,578					5,840				
p-value	0.000					0.000					0.000				

Note: All regressions include a set of biannual calendar time dummy variables (not reported).
a. Percentage change in the hazard ratio if the value of the exogenous variables increases from the 25th percentile to the 75th percentile of all observations in the dataset. In the case of dummy variables, the impact of a change from 0 to 1 is reported.
Source: Regression results based on data from UN Comtrade.

However, the small number of low-income country importers in the sample requires caution in interpreting this result.

IV. DISCUSSION AND CONCLUSIONS

Exporting is a perilous activity, particularly in low-income countries. Understanding the determinants of export survival rates is therefore an important challenge for empirical analysis. A policy focus only on entry into exporting will miss a fundamental aspect of the dynamics of exporting. A strategy for increasing and sustaining export growth rates should address factors that contribute to the low export survival rates.

This article extends the path-breaking studies of export survival by Besedes and Prusa (2006a, b) by applying statistical survival techniques to a broad dataset of bilateral export flows with a high level of product and export-country detail. It formally tests, and rejects, the assumption of proportional hazards that underlies the standard Cox (1972) model typically used in the literature. The finding of the importance of accounting for unobserved heterogeneity points to the need for survival analysis with firm-level data to complement studies using product-level flow data in modeling export survival. The study also finds that both regional- and product-specific experience matter most for low-income economies and that these countries appear to insufficiently exploit the advantages of trading with neighboring economies.

The aim of this article was to extend the empirical analysis of export survival. There is still a way to go before clear policy recommendations can be derived, however. That will require more detailed country- and firm-specific analysis of the factors influencing export survival. The high prevalence of export failures suggests some caution regarding policy messages about constraints to exporting drawn from surveys of existing exporters. It is also important to derive information from exporting firms that have failed.

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APPENDIX. SOURCES AND DEFINITIONS OF VARIABLES

Appendix 1. Variable sources and definitions

Variable	Definition	Source
Initial value	Log of the trade value in first year of spell	UN Comtrade
Distance	Log of the distance in kilometers between most populous cities in exporting and importing countries	CEPII
Border	Dummy variable, 1 if shared border	CEPII
Common language	Dummy variable, 1 if common language	CEPII
Colony	Dummy variable, 1 if exporter is a former colony of importer	CEPII
GDP Imp x Exp	Product of GDP of importer and exporter in purchasing power parity terms	IMF
Total exports	Log of total exports of the exporter	UN Comtrade
Industry exports	Log of exporter's total exports in industry	UN Comtrade
Product exports	Log of exports of the product to all countries by the exporter in first year of trade relationship	UN Comtrade
Product imports	Log of imports of the product from all countries by the importer in the first year of trade relationship	UN Comtrade
Neighbor	Dummy variable, 1 if exporter exports the same product to a neighboring country	UN Comtrade
Bilateral trade	Log of exports of all products from exporter to importer in the first year of trade relationship	UN Comtrade
Previous spell	Dummy variable, 1 if previous spells within sample period	UN Comtrade
FX volatility	Average absolute value of the monthly percentage change in the exchange rate between exporter and importer in years $t = -1, 0$ and 1 around the beginning of the trade relationship	IMF
FX Misalignment	Exchange rate between exporter and importer in the year the trade relationship starts relative to the period average.	IMF
PTA	Dummy variable, 1 if exporter and importer are in a preferential trade agreement when trade flow commences	World Bank